

Combining Function Types

Functions can be combined using addition, subtraction, multiplication, and division.

Definitions

Functions can be combined using the four arithmetic operations.

Function Addition $(f + g)(x) = f(x) + g(x)$

Function Subtraction $(f - g)(x) = f(x) - g(x)$

Function Multiplication $(fg)(x) = f(x) \cdot g(x)$

Function Division $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$

EXAMPLE 1

Write a rule for the indicated combination of the functions

$f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$.

A $(f + g)(x)$

SOLUTION

$$(f + g)(x) = f(x) + g(x)$$

$$= (x^2 - 2x - 3) + (x + 1) \quad \text{Substitute the rules.}$$

$$= x^2 - 2x + x - 3 + 1 \quad \text{Commutative Property}$$

$$= x^2 - x - 2 \quad \text{Simplify.}$$

The rule is $(f + g)(x) = x^2 - x - 2$.

B $(f - g)(x)$

SOLUTION

$$\begin{aligned}(f - g)(x) &= f(x) - g(x) \\ &= (x^2 - 2x - 3) - (x + 1) && \text{Substitute the rules.} \\ &= (x^2 - 2x - 3) - x - 1 && \text{Distributive Property} \\ &= x^2 - 2x - x - 3 - 1 && \text{Commutative Property} \\ &= x^2 - 3x - 4 && \text{Simplify.}\end{aligned}$$

The rule is $(f - g)(x) = x^2 - 3x - 4$.

C $(fg)(x) = f(x) \cdot g(x)$

SOLUTION

$$\begin{aligned}(fg)(x) &= f(x) \cdot g(x) \\ &= (x^2 - 2x - 3)(x + 1) && \text{Substitute the rules.} \\ &= (x^3 - 2x^2 - 3x) + (x^2 - 2x - 3) && \text{Distributive Property} \\ &= x^3 - x^2 - 5x - 3 && \text{Simplify.}\end{aligned}$$

The rule is $(fg)(x) = x^3 - x^2 - 5x - 3$.

D $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$

SOLUTION

$$\begin{aligned}\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)} \\ &= \frac{x^2 - 2x - 3}{x + 1} && \text{Substitute the rules.} \\ &= \frac{(x - 3)(x + 1)}{x + 1} && \text{Factor.} \\ &= x - 3 && \text{Divide out the common factor.}\end{aligned}$$

Because $g(x)$ cannot be zero, x cannot be -1 .

The rule is $\left(\frac{f}{g}\right)(x) = x - 3, x \neq -1$. ■

Evaluating Combined Functions

You can evaluate combined functions for specific values of x .

EXAMPLE 2

Given $f(x) = 2x - 3$ and $g(x) = 5 - x$, evaluate the function expression.

A $(f - g)(4)$

SOLUTION

$$(f - g)(4) = f(4) - g(4)$$

$$= (2 \cdot 4 - 3) - (5 - 4) \quad \text{Substitute 4 for } x \text{ in the rules.}$$

$$= (8 - 3) - 1 \quad \text{Use the order of operations.}$$

$$= 4 \quad \text{Simplify.}$$

B $\left(\frac{g}{f}\right)(-1)$

SOLUTION

$$\left(\frac{g}{f}\right)(-1) = \frac{g(-1)}{f(-1)}$$

$$= \frac{5 - (-1)}{2(-1) - 3} \quad \text{Substitute } -1 \text{ for } x \text{ in the rules.}$$

$$= -\frac{6}{5} \quad \text{Simplify. } \blacksquare$$